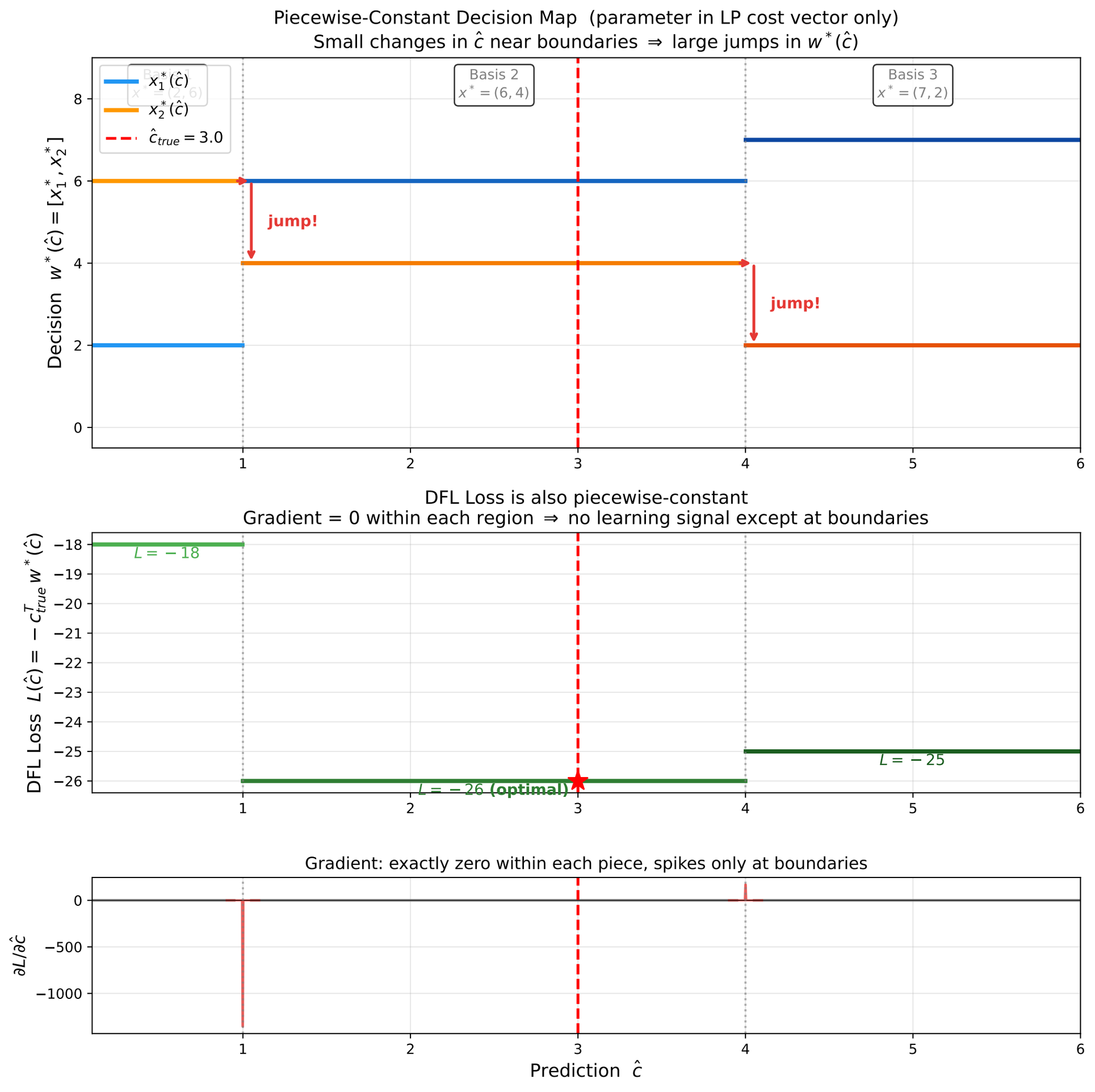
Piecewise-Constant Decision Map (parameter in LP cost vector only)
Small changes in $\hat{c}$ near boundaries ⇒ large jumps in $w^*(\hat{c})$
Basis 2
$x^* = (6, 4)$
Basis 3
$x^* = (7, 2)$
$x_1^*(\hat{c})$
$x_2^*(\hat{c})$
$\hat{c}_{true} = 3.0$
jump!
jump!
Decision $w^*(\hat{c}) = [x_1^*, x_2^*]$
DFL Loss is also piecewise-constant
Gradient = 0 within each region ⇒ no learning signal except at boundaries
$L = -18$
$L = -26$ (optimal)
$L = -25$
DFL Loss $L(\hat{c}) = -c_{true}^T w^*(\hat{c})$
Gradient: exactly zero within each piece, spikes only at boundaries
$\partial L/\partial \hat{c}$
Prediction $\hat{c}$